

Rémy Hosseinkhan-Boucher

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Experience

- 2025-2027 **Centre de Mathématiques Appliquées (CMAP), École Polytechnique** - Palaiseau, France
Postdoctoral Researcher
Learning closures for hyperbolic systems in physics, in particular moment closures of kinetic equations. Constrained learning for financial PDEs (terminal payoff, boundary conditions). Research teams: HPC@Maths, MathsFi.
Collaborators: Marc Massot, Charles-Albert Lehalle
- 2021–2025 **Laboratory of Interdisciplinary Digital Sciences (LISN), Université Paris-Saclay** - Orsay, France
Ph. D. Candidate (Full-time): On Learning-Based Control of Dynamical Systems.
Activities: Research, software development, scientific communication, teaching, reviewing, internship supervision.
Conference publications. Research groups: DATAFLOT; Learning & Optimisation (A&O); Inria TAU.
Advisors: A. Vilnat, O. Semeraro, L. Mathelin.
- 2020–2021 **Inria TAU (TACKling the Underspecified), IFP Énergies Nouvelles** - Gif-sur-Yvette, France
Research Intern
Learning-based methods for solving stiff differential equations. Koopman operator theory and Physics Informed Neural Networks (PINNs).
Advisors: Michele Alessandro Bucci, Thibault Faney, Cédric Mehl
- 2019–2020 **BNP Paribas Real Estate** - Issy-les-Moulineaux, France
Data Scientist
Cluster analysis using k-means and Gaussian mixtures. Temporal analysis of clusters inspired by NLP-based dynamic word embeddings. Exploratory data analysis, feature engineering, and variable selection with ANOVA and PCA.
- 2019 **Capital Fund Management (CFM)** - Paris, France
Quantitative Research Intern
Anomaly detection methods for financial time series. Modelling with ARIMA, GARCH, and Facebook PROPHET. Parameter estimation of heavy-tailed distributions (Levy).
- 2018 **Luxuryinsight** - Paris, France
Deep Learning Research Intern
Deep learning models for Natural Language Processing (NLP) tasks with RNN, GRU, LSTM neural networks. Embedding techniques and multi-class classification.

Education

- 2021–2024 **Université Paris-Saclay**
Ph.D. in Computer Science
On Learning-Based Control of Dynamical Systems

Advisors: Anne Vilnat, Onofrio Semeraro, Lionel Mathelin

2014–2020

Université Paris Dauphine - PSL

M.Sc. in Artificial Intelligence, Systems, Data

Statistical Learning, Markov Decision Processes, Parallel Computing

M.Sc. in Statistics and Data Analysis for Financial Engineering

Time Series Analysis, Stochastic Calculus, Nonparametric Statistics

B.Sc. in Applied Mathematics

Advanced Probability Theory, Stochastic Processes, Parametric / Bayesian Statistics

Publications



Peer-reviewed Conference Proceedings

- C1. Pradeleix, E., **Hosseinkhan-Boucher, Rémy**, Shilova, A., Semeraro, O. & Mathelin, L. *Learning non-Markovian Dynamical Systems with Signature-based Encoders in Proceedings of the 2nd ECAI Workshop on "Machine Learning Meets Differential Equations: From Theory to Applications"* (eds Coelho, C., Zimmering, B., Costa, M. F. P., Ferrás, L. L. & Niggemann, O.) **277** (PMLR, 26 Oct 2025), 1–25. <https://proceedings.mlr.press/v277/pradeleix25a.html>.
- C2. **Hosseinkhan-Boucher, Rémy**, Douka, S., Semeraro, O. & Mathelin, L. *Increasing information for model predictive control with semi-Markov decision processes in Proceedings of the 6th Annual Learning for Dynamics & Control Conference* (eds Abate, A., Cannon, M., Margellos, K. & Papachristodoulou, A.) **242** (PMLR, July 2024), 1400–1414. <https://proceedings.mlr.press/v242/hosseinkhan-boucher24a.html>.
- C3. **Hosseinkhan-Boucher, Rémy**, Semeraro, O. & Mathelin, L. *Evidence on the Regularisation Properties of Maximum-Entropy Reinforcement Learning in Proceedings of the 7th International Conference on Optimization and Learning* (eds Dorronsoro, B., Chicano, F., Danoy, G. & Talbi, E.-G.) (May 2024).

Working papers

- W1. **Hosseinkhan-Boucher, Rémy**, Biasone, D. & Massot, M. *Learning the Regularisation of a Kinetic 5-Moment Closure* In preparation. 2026.
- W2. **Hosseinkhan-Boucher, Rémy** & Lehalle, C.-A. *Exact Terminal-Condition Neural Networks for Contingent-Claim Valuation under Non-Smooth Payoffs* In preparation. 2026.
- W3. **Hosseinkhan-Boucher, Rémy**, Monsel, T., Semeraro, O. & Mathelin, L. *Continuous-Time Reinforcement Learning: Modeling Delayed Dynamics with Neural Delay Differential Equations* 2025.

Presentations

Talks

- T1. **Hosseinkhan-Boucher, Rémy**. *Learning the Regularisation of a Kinetic 5-Moment Closure* Minisymposium on machine learning for PDEs (self-organised), Congrès National d'Analyse Numérique (CANUM 2026), Saint-Jacut-de-la-Mer, France. June 2026.
- T2. **Hosseinkhan-Boucher, Rémy**. *Challenges in Learning Based Control for Dynamical Systems: Maximum Entropy and Mutual Information* Inria TAU Seminar Series. Feb. 2024.
- T3. **Hosseinkhan-Boucher, Rémy**. *Gaussian Process Regression on Vector Fields and Uncertainty Quantification* Mechanics Department Seminar, LISN, Orsay, France. 2024.
- T4. **Hosseinkhan-Boucher, Rémy**. *Robustness of Maximum-Entropy Reinforcement Learning* SIAM Conference on Science and Engineering. Feb. 2023.

- T5. **Hosseinkhan-Boucher, Rémy.** *A Reinforcement Learning Application to Chaotic Dynamical Systems* European Drag Reduction and Flow Control Meeting. Sept. 2022.

Teaching

École Polytechnique

- 2026 *Teaching Assistant, Machine Learning for Scientific Computing and Numerical Analysis*
Joint course with TU Eindhoven (EuroTeQ alliance).
Supervise practical sessions and final exam.
Principal Lecturers: H. Montanelli, V. Dolean.

ENS Paris-Saclay

- 2024 *Teaching Assistant, Advanced Deep Learning (MVA Programme)*
Advanced deep learning concepts to MVA (Math Vision Apprentissage) Master's students.
Machine Learning for Physics and Computer Vision.
Principal Lecturer: G. Charpiat.

Université Paris-Saclay

- 2025-2026 *Teaching Assistant, Reinforcement Learning (M2 Artificial Intelligence)*
Dynamic Programming, Stochastic Approximation, and Function Approximation methods; designed and supervised practical sessions and final exam.
Principal Lecturer: L. Mathelin.
- 2022–2023 *Teaching Assistant, C++ Programming*
Object-oriented programming (OOP) to 1st-year students.
Principal Lecturers: C. Balkanski, H. Bonneau.

CentraleSupélec

- 2026 *Teaching Assistant, Data Science Project Class*
Guided 1st-year students in their data science project on deep learning for soil image classification (vineyard application).
Principal Lecturer: W. Ouerdane.
- 2022 *Teaching Assistant, Data Science Project Class*
Guided 2nd-year students in their data science project on adversarial robustness for deep learning classifiers.
Partnership with IRT SystemX. Principal Lecturer: W. Ouerdane.

Research Supervision

Internships

- 2026 *Salma Ouaisi, ENSIMAG (Grenoble INP – Université Grenoble Alpes)*
Topic: Constrained Learning for Financial Partial Differential Equations: Exact-Constraint Ansatz.
Role: Co-advisor with C.-A. Lehalle.
- 2025–26 *Joachim Jobard, École Centrale Lyon KTH Royal Institute of Technology*
Topic: Learning-based Dynamic Programming on Functional Differential Equations.
Role: Co-advisor with L. Mathelin and O. Semeraro.
- 2025 *Elliott Pradeleix, École Polytechnique*

Topic: Learning Functional Differential Equations with Signature-based Encoders.
Led to a publication in the Proceedings of Machine Learning Research (PMLR).
Role: Co-advisor with A. Shilova, L. Mathelin and O. Semeraro.

2023 *Stella Douka, Université Paris-Saclay, M.Sc. in Artificial Intelligence*
Topic: Gaussian Process based Model Predictive Control with Mutual Information criterion.
Led to a publication in the Proceedings of Machine Learning Research (PMLR).
Role: Co-advisor with L. Mathelin and O. Semeraro

Academic Service

Conference Organisation

2026 **Organiser (lead)**, minisymposium “*Perspectives et avancées récentes en apprentissage pour la résolution numérique d’équations aux dérivées partielles*”, Congrès National d’Analyse Numérique (CANUM 2026), Saint-Jacut-de-la-Mer, France. Convened six young researchers (doctoral and post-doctoral) on machine learning for the numerical solution of partial differential equations.

Peer Review

2025 International Conference on Machine Learning (ICML)
2024 Journal of Fluid Mechanics (JFM)
2024 European Workshop on Reinforcement Learning (EWRL)
2023 IEEE Transactions on Automatic Control (TAC)

Tools & Software

Library

`control_dde`: Learning-based control for delay-differential systems.

Skills

Programming: Python, C++, R, Java, Scala, Bash
Tools: Git, Docker, Singularity, MLFlow, Hydra, Slurm
Frameworks: TensorFlow, PyTorch, Spark, Hadoop
Other: LaTeX, Markdown, Jupyter

Other Experience

2015 **Europ Assistance**
Assistance Agent
Processed customer calls based on individual insurance contracts.
Managed correction and tracking of cases until closure, transmitting data to partners.

2014 **Hilton**
Room Attendant at Day’s Inn
Worked in Florida for 3 months as a housekeeper and breakfast waiter, enhancing English proficiency and adapting to new cultures (Visa J1).